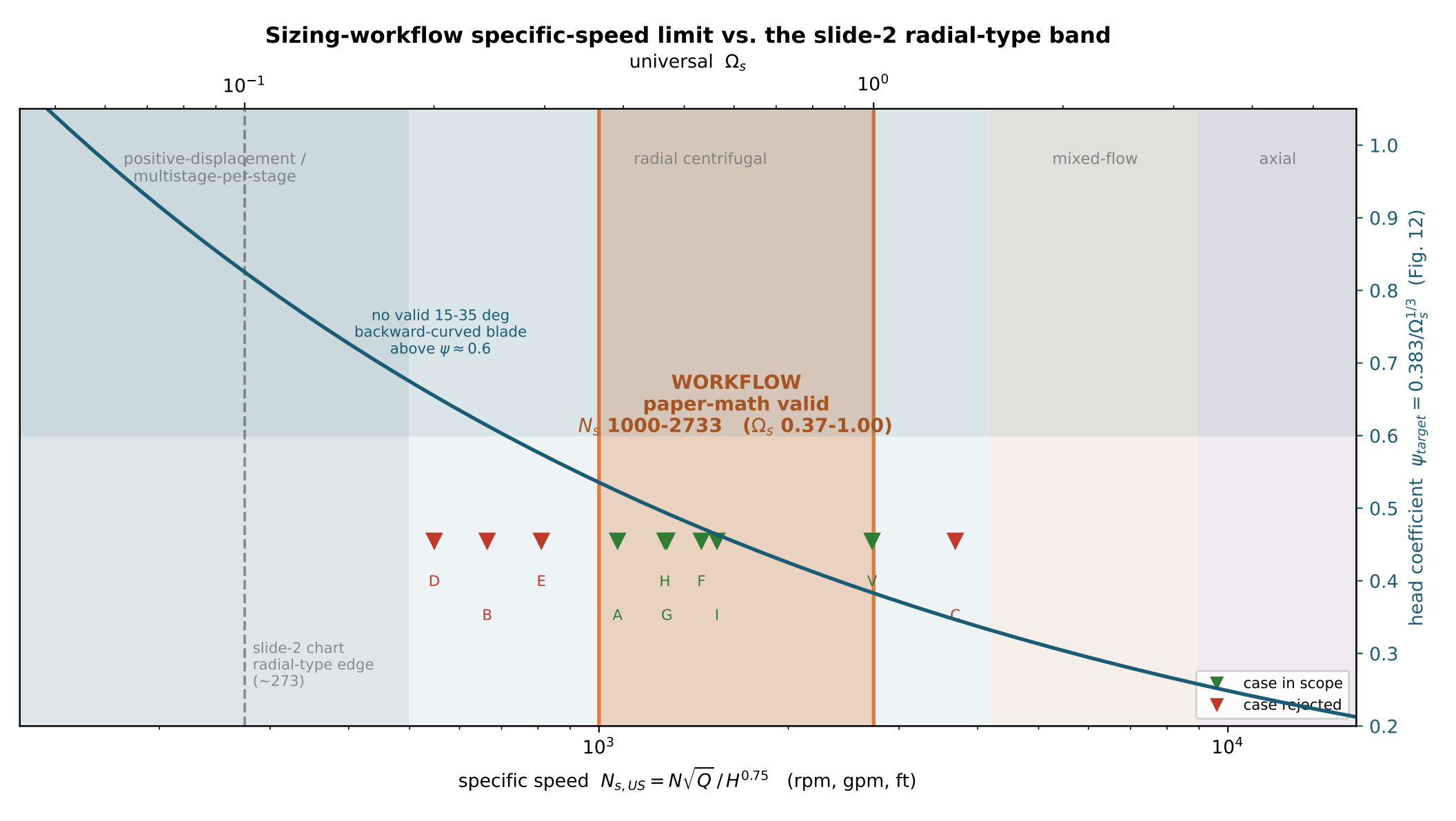
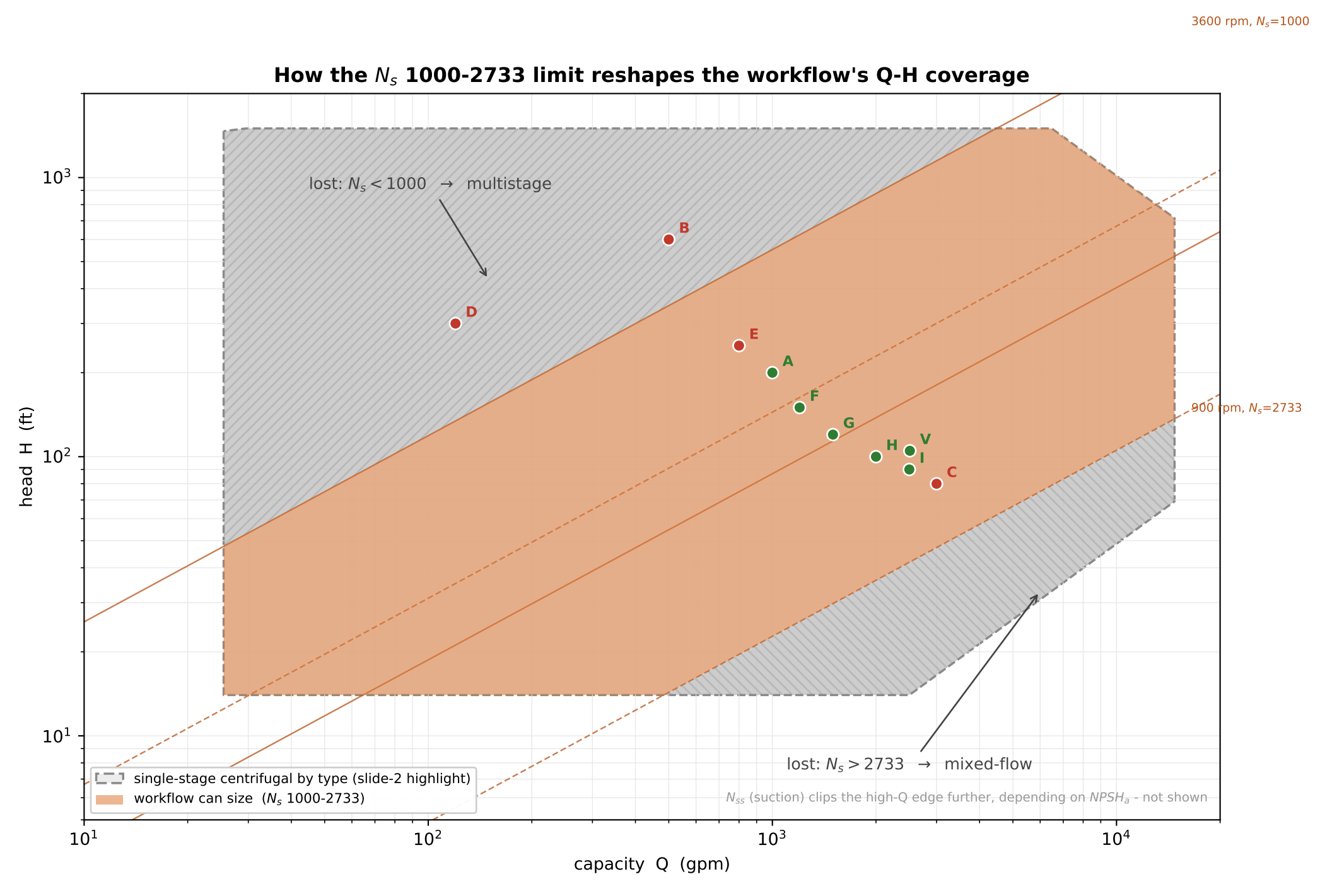

Sizing-workflow specific-speed limit vs. the slide-2 radial-type band
universal Ω_s
10^{-1}
10^{0}
positive-displacement / multistage-per-stage
radial centrifugal
mixed-flow
axial
no valid 15-35 deg backward-curved blade above $\psi \approx 0.6$
WORKFLOW
paper-math valid
N_s 1000-2733 (Ω_s 0.37-1.00)
D
B
E
A
H
G
F
I
V
C
slide-2 chart radial-type edge (~273)
case in scope
case rejected
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
head coefficient $\psi_{target} = 0.383/\Omega_s^{1/3}$ (Fig. 12)
10^{3}
10^{4}
specific speed $N_{s,US} = N\sqrt{Q}\,/\,H^{0.75}$ (rpm, gpm, ft)


How the N_s 1000-2733 limit reshapes the workflow's Q-H coverage
3600 rpm, N_s=1000
900 rpm, N_s=2733
lost: $N_s < 1000$ → multistage
lost: $N_s > 2733$ → mixed-flow
N_{ss} (suction) clips the high-Q edge further, depending on $NPSH_a$ - not shown
single-stage centrifugal by type (slide-2 highlight)
workflow can size (N_s 1000-2733)
B
D
E
A
F
G
H
V
I
C
10^3
10^2
10^1
head H (ft)
10^1
10^2
10^3
10^4
capacity Q (gpm)